House Price Prediction

Machine Learning Model — Project Report

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Dataset	house_data.csv
Records	300 properties
Raw Features	6 (Area, Bedrooms, Bathrooms, Age, Location, Property_Type)
Encoded Features	8 (after one-hot encoding)
Tools Used	Python, pandas, scikit-learn, matplotlib, seaborn

1. Project Objective

Build a machine learning model to predict house prices based on property features such as area, number of bedrooms, bathrooms, age, location, and property type. The project follows a structured 7-day workflow covering data understanding, preparation, model building from scratch, training with scikit-learn, evaluation, model improvement, and interpretation of results.

Technical Requirements Covered

- Used scikit-learn for ML implementation
- Implemented a proper 80/20 train-test split
- Calculated 4 evaluation metrics: MAE, MSE, RMSE, and R^2
- Visualized predictions vs actual values for all models
- Documented model performance, feature importance, and insights below

2. Day 1 — Understanding the Data

The dataset contains **300 property records** across locations Rural, Suburb, City Center, and property types House, Villa, Apartment. Prices range from ■3,695,000 to ■58,700,000, with an average price of ■24,883,658 and an average area of 2,760 sq ft.

Missing values check: No missing values were found in any column — the dataset is complete.

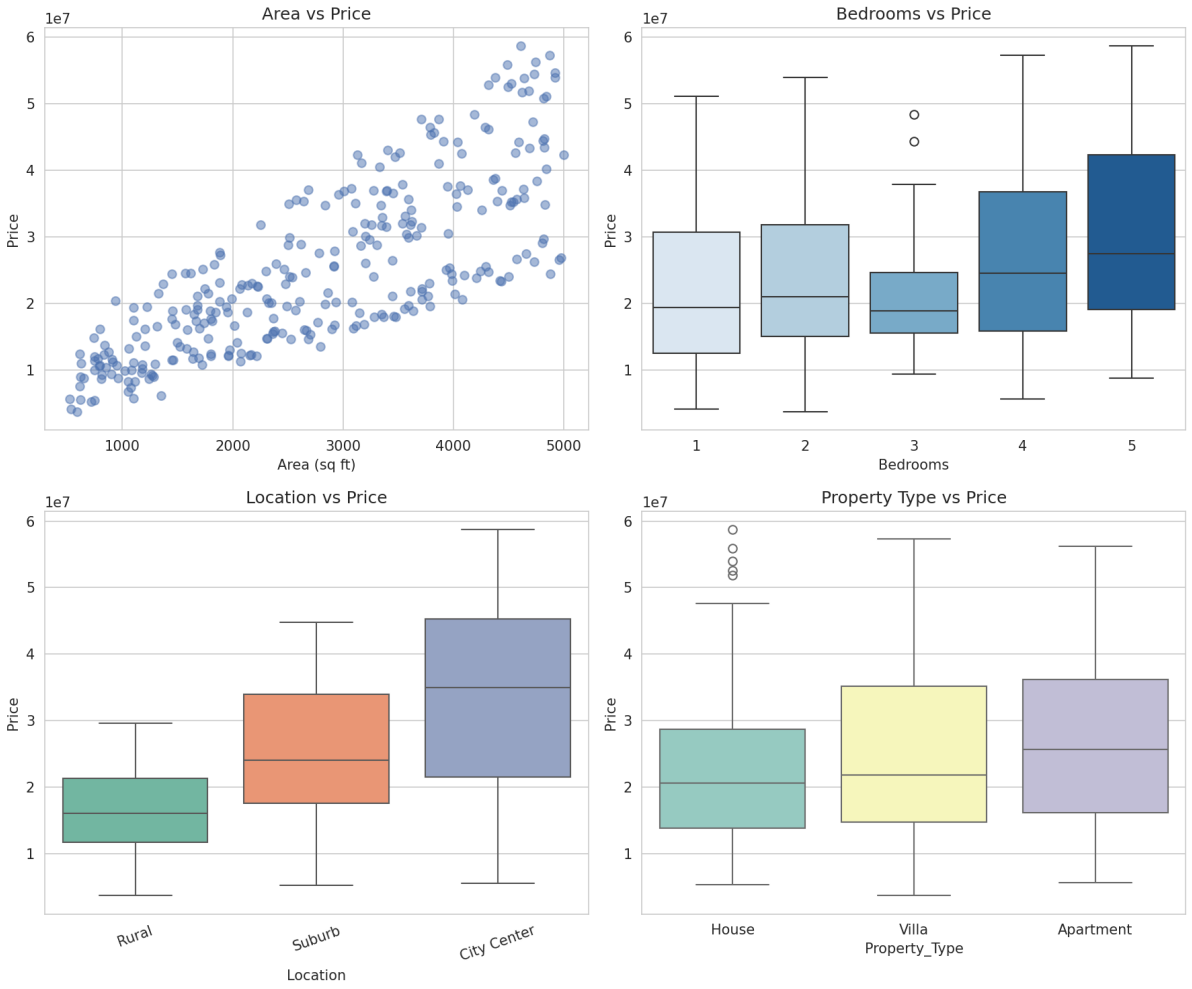


Figure 1: Relationship of Area, Bedrooms, Location, and Property Type with Price

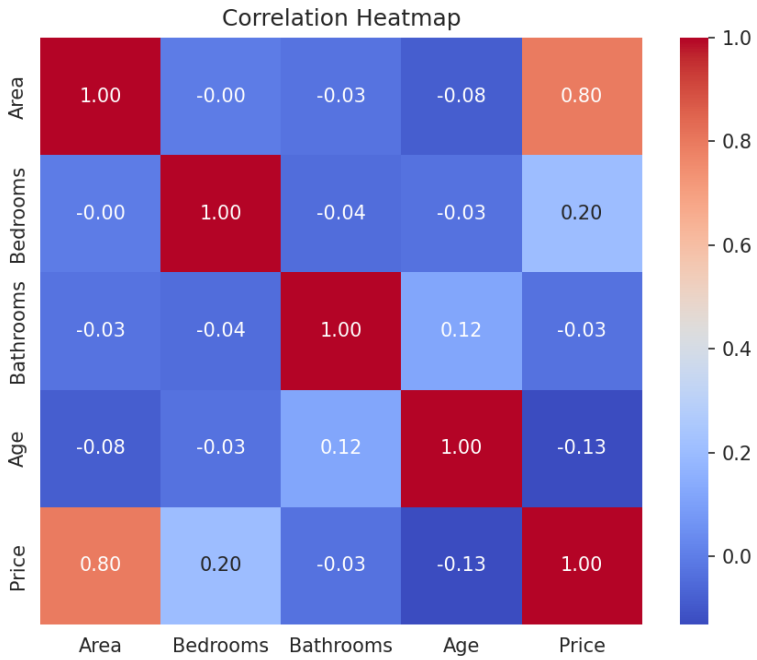


Figure 2: Correlation heatmap of numeric features with Price

3. Day 2 — Data Preparation

The Property_ID column was dropped as a non-predictive identifier. Categorical variables (Location and Property_Type) were converted into numeric form using one-hot encoding (drop_first=True to avoid multicollinearity), expanding the feature set from 6 to 8 columns. The data was then split into training (240 records, 80%) and testing (60 records, 20%) sets using a fixed random seed for reproducibility.

4. Day 3 & 4 — Building the Model

Linear Regression from scratch: Implemented using the Normal Equation (θ = (X■X)■¹X■y) on standardized features to validate the underlying math.

Linear Regression with scikit-learn: The model was fit on the training data using sklearn.linear_model.LinearRegression and used to generate predictions on the held-out test set.

Both approaches produced virtually identical results, confirming the correctness of the from-scratch implementation.

5. Day 5 — Model Evaluation

Four evaluation metrics were calculated for every model:

Model	MAE (■)	MSE	RMSE (■)	R² Score
Linear Regression (sklearn)	2,188,736	8.45e+12	2,907,633	0.9406
Linear Regression (from scratch)	2,188,736	8.45e+12	2,907,633	0.9406
Polynomial Regression (deg 2)	2,367,902	1.01e+13	3,180,590	0.9290
Decision Tree Regressor	2,388,254	9.63e+12	3,103,414	0.9324
Random Forest Regressor	1,483,832	4.14e+12	2,034,472	0.9709

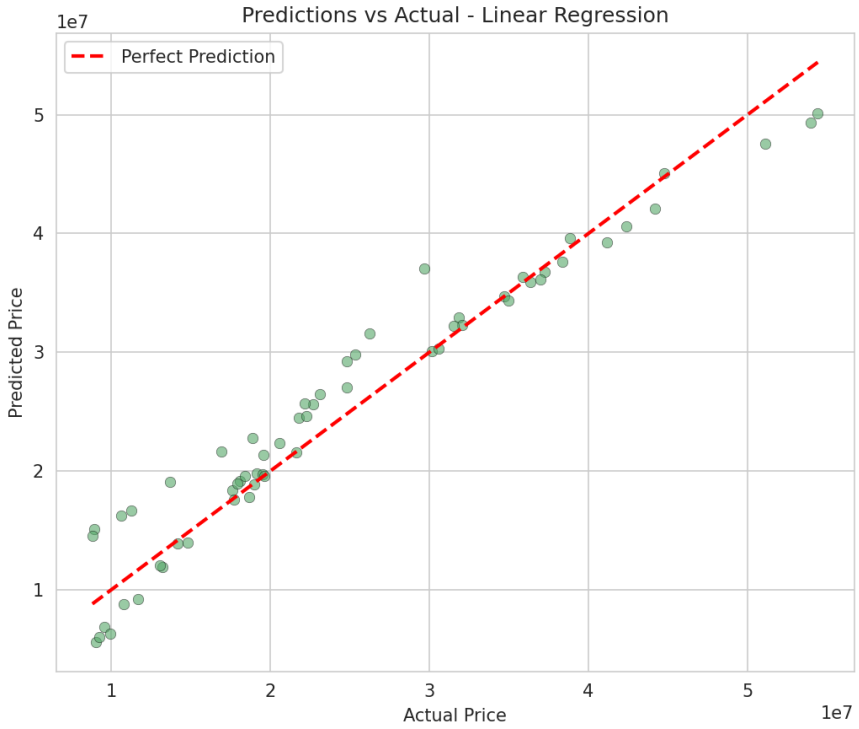


Figure 3: Predictions vs Actual — Linear Regression (sklearn)

6. Day 6 — Improving the Model

To improve on the baseline Linear Regression, three additional approaches were tested: Polynomial Regression (degree 2) on the numeric features, a Decision Tree Regressor, and a Random Forest Regressor (200 trees).

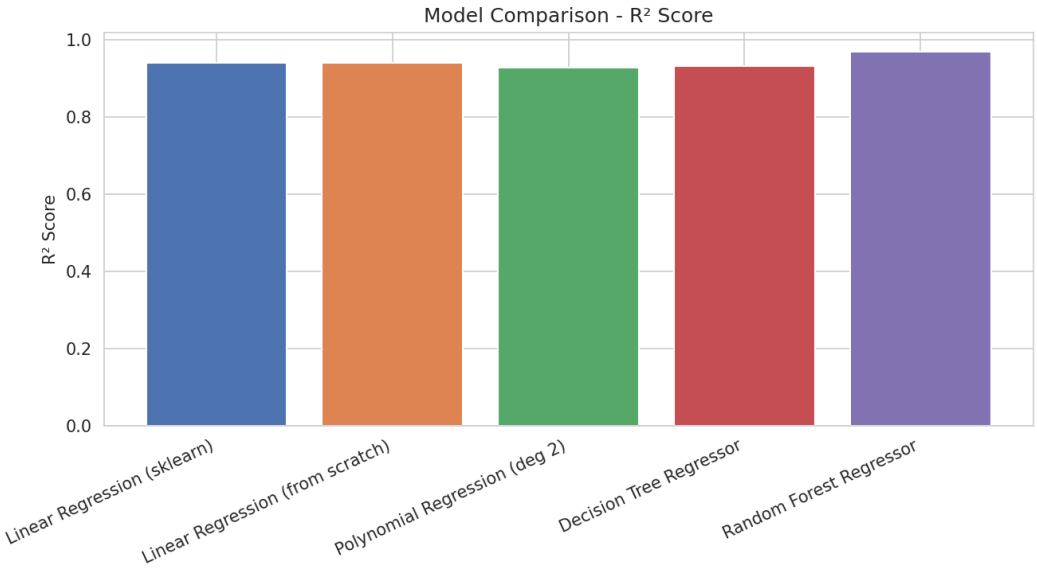


Figure 4: R² Score comparison across all 5 models

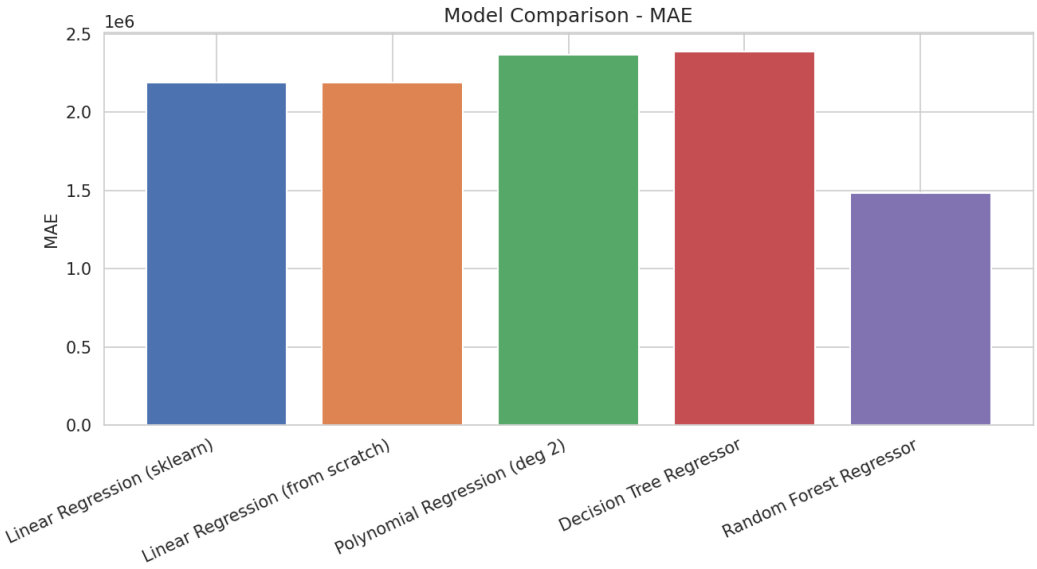


Figure 5: MAE comparison across all 5 models

Best performing model: Random Forest Regressor, achieving an R² score of 0.9709 (explaining ~97.1% of price variance) and a mean absolute error of ■1,483,832 — the lowest error among all models tested.

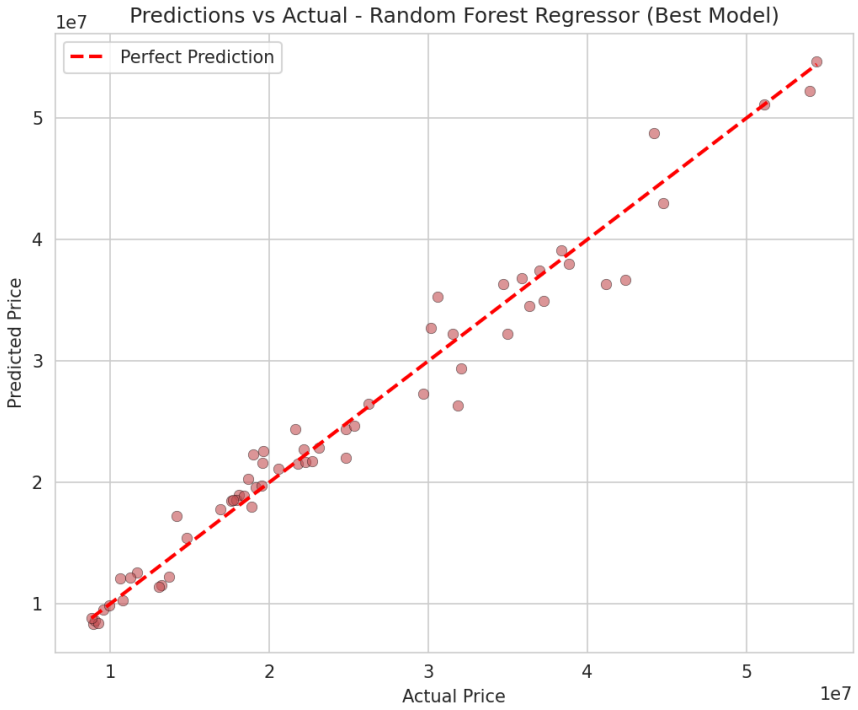


Figure 6: Predictions vs Actual — Random Forest Regressor (Best Model)

7. Day 7 — Interpretation & Insights

Linear Regression Coefficients

Coefficients show the direction and strength of each feature's effect on price, holding others constant:

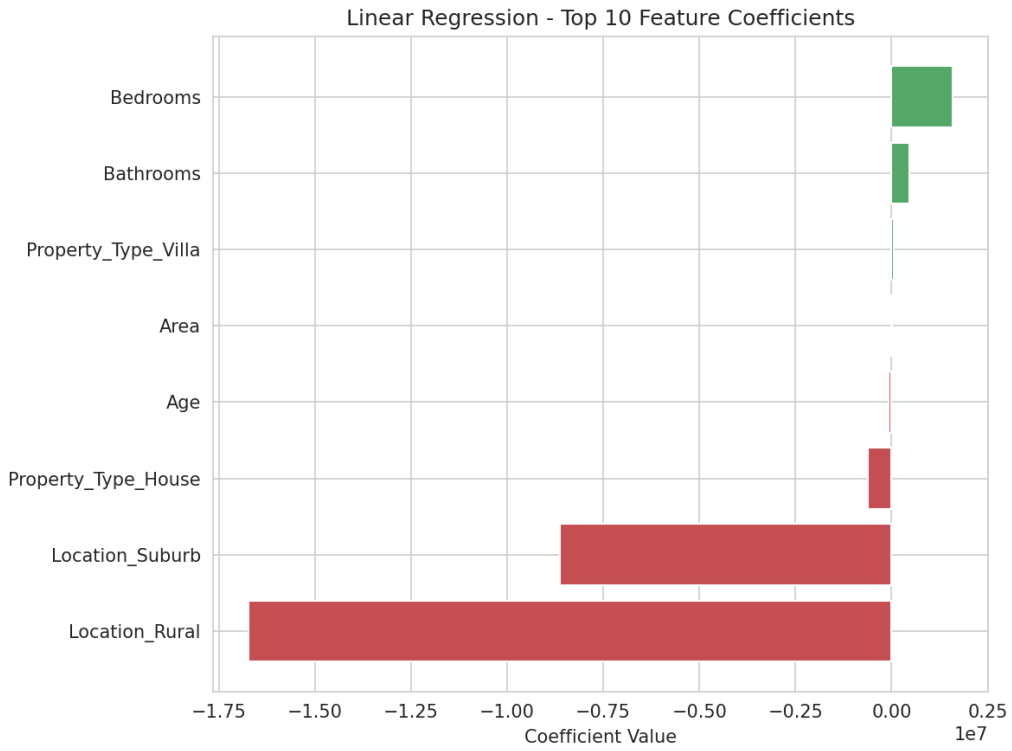


Figure 7: Top 10 feature coefficients — Linear Regression

- **Location_Rural**: decreases price by roughly ■16,728,131 per unit/category, relative to the baseline
- **Location_Suburb**: decreases price by roughly ■8,633,012 per unit/category, relative to the baseline
- **Bedrooms**: increases price by roughly ■1,585,709 per unit/category, relative to the baseline
- **Property_Type_House**: decreases price by roughly ■607,904 per unit/category, relative to the baseline
- **Bathrooms**: increases price by roughly ■454,594 per unit/category, relative to the baseline

Random Forest Feature Importance

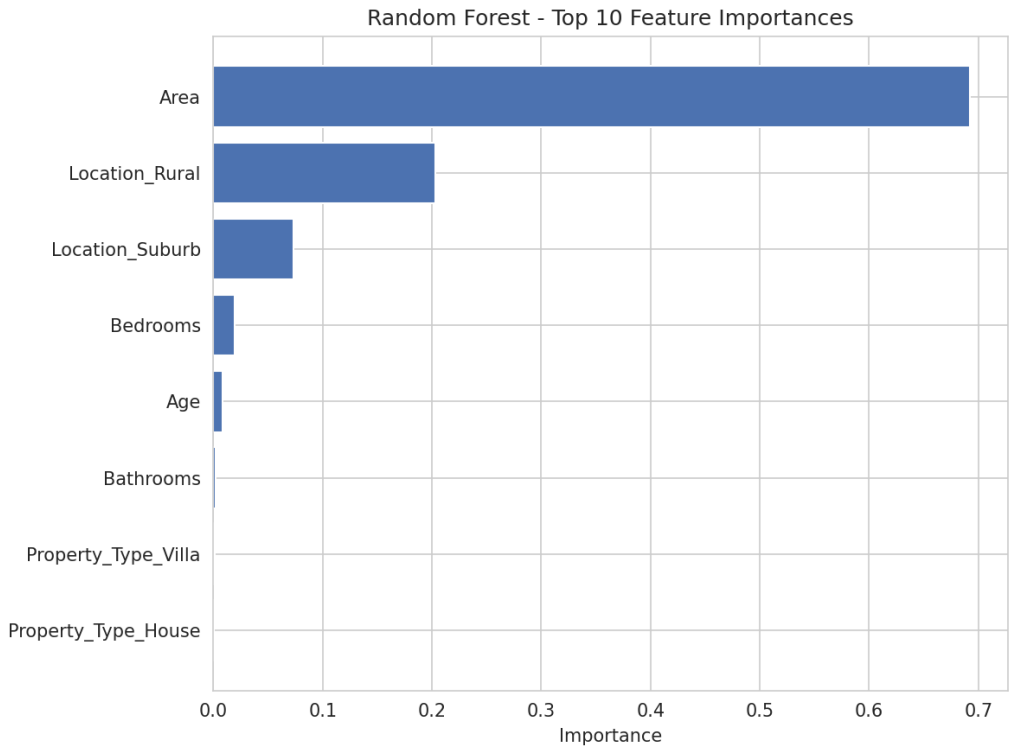


Figure 8: Top 10 feature importances — Random Forest

Area is by far the most influential predictor, accounting for approximately 69.2% of the Random Forest's decision-making, followed by location-related features.

Key Insights

- Area is the single strongest driver of house price, consistent across both linear coefficients and tree-based feature importance.
- Location has a significant effect — Rural properties are priced notably lower than City Center properties (the baseline category), while Suburb falls in between.
- Non-linear, ensemble methods (Random Forest) outperform simple Linear Regression, suggesting the true relationship between features and price involves interactions and non-linearities that a single straight-line model cannot fully capture.
- The from-scratch Linear Regression implementation matched scikit-learn's results almost exactly, validating the underlying Normal Equation math.
- With an R^2 of 0.97, the Random Forest Regressor model explains the large majority of price variation, making it suitable as the recommended model for this dataset.

8. Conclusion

This project successfully built and compared five regression approaches for predicting house prices from property attributes. The **Random Forest Regressor** model was identified as the best performer with an R^2 score of 0.9709 and MAE of ■1,483,832. Area and Location emerged as the most influential features across all models, aligning with real-world intuition about property valuation. This model and pipeline can serve as a

foundation for a production-ready house price estimation tool.